Course 1 Module 2

Data Mining Process

- Establishing Data Mining Goals: Define objectives, consider cost-benefit trade-offs, and determine accuracy level.
- Selecting Data: Choose quality data, identify sources, and consider cost of collection.
- Preprocessing Data: Clean messy data, remove errors, and handle missing values.
- Transforming Data: Convert variables, reduce dimensions, and aggregate information.
- Storing Data: Store securely with read/write access and organized structure.
- Mining Data: Apply algorithms and visualization techniques.
- Evaluating Results: Test models, gather feedback, and iterate.

Regression Concept

- Tall parents usually have tall children but not necessarily taller than themselves.
- Regression was introduced by Sir Francis Galton.
- Regression models are widely used in business, medicine, and research.
- Regression measures the magnitude of relationships.
- Example: Additional washroom adds more value than an additional bedroom.
- Distance to subway increases price; proximity to highway reduces price.
- Regression quantifies relationships for decision making.

Key Concepts Summary

- Big Data has five characteristics: Velocity, Volume, Variety, Veracity, Value.
- Cloud computing characteristics: On-demand self-service, Broad network access, Resource pooling, Rapid elasticity, Measured service.
- Data mining has six steps: Goal setting, Selecting data, Preprocessing, Transforming, Mining, Evaluation.
- Large data requires scalable and innovative technologies.
- Deep learning uses neural networks to identify patterns.
- Machine learning uses algorithms to learn and make predictions.
- Regression identifies relationships between variables.
- Big Data skills include statistics, machine learning, and programming.
- Generative AI creates new content like text, images, music, and code.